STATUTORY REQUIREMENT (Part-ADR.OR.D.005 • Finding Remediation): Under Regulation (EU) No 139/2014, when a competent civil aviation authority issues an audit or inspection finding to an aerodrome certificate holder, the operator is legally required to identify the root cause, formulate a formal Corrective Action Plan (CAP), establish interim safety mitigations, and obtain authority concurrence within defined statutory timelines.

1. LEGAL TAXONOMY OF AUDIT FINDINGS

LEVEL 1 FINDING • SIGNIFICANT SAFETY HAZARD

Any significant non-compliance with aerodrome certification specifications, the aerodrome manual, or operational requirements which lowers safety or seriously hazards aircraft operations.

Authority Action: Immediate restriction, partial closure, or certificate limitation.
Remediation Timeline: Immediate provisional mitigation (max 24–48 hours).

LEVEL 2 FINDING • NON-IMMEDIATE COMPLIANCE GAP

Any non-compliance with aerodrome regulations or operating procedures which could lower safety or hazard aircraft operations if not corrected within an agreed timeframe.

Authority Action: Formal notification requiring CAP submission.
Remediation Timeline: CAP submitted within 30 days; full closure within 3 months.

2. EASA CAP CORE STRUCTURAL COMPONENTS

COMPONENT	REGULATORY INTENT	SPOKEN ENGLISH & OPERATIONAL IMPLEMENTATION
1. Root-Cause Analysis	Identify systemic origin (James Reason model).	Differentiate between immediate technical symptoms and organizational root causes.
2. Immediate Mitigation	Contain operational risk pending permanent works.	Issue temporary NOTAMs, operational weight limits, or progressive taxi clearances.
3. Permanent Action Plan	Definitive engineering or procedural fix.	Capital works, equipment procurement, contractor bidding, and SOP revision.
4. Verification Milestones	Auditable check-points with target dates.	Progress benchmarks submitted to the competent authority for verification.
5. Authority Sign-Off	Formal closure notice under ADR.AR.C.055.	Authority on-site physical re-inspection and written letter of finding closure.

3. SPOKEN NEGOTIATION FORMULAS FOR AUTHORITY AUDITS

AERODROME OPERATOR (COMMITMENT & ASSURANCE)

- *"We undertake to implement the interim containment barrier within 14 days."*
- *"Pending permanent re-surfacing, a NOTAM shall restrict wet operations."*
- *"We propose milestone verification audits on the 15th of each month."*

AUTHORITY INSPECTOR (SCRUTINY & ACCEPTANCE)

- *"The proposed timeline for civil works is acceptable, subject to monthly reporting."*
- *"Level 1 finding closure remains conditional upon physical on-site friction testing."*
- *"The aerodrome operator must ensure wing-walker procedures are fully reinforced."*

FINDING STATEMENT • RUNWAY 14L FRICTION DECAY LEVEL 1 NON-COMPLIANCE

Continuous Friction Measuring Equipment (CFME) survey conducted on 08 October 2024 revealed that the runway surface friction coefficient within the first 600m of Runway 14L touchdown zone has decayed to **μ = 0.31** at 65 km/h with a 1.0mm water film. This value is critically below the EASA Part-ADR.OPS.B.015 Minimum Friction Level (μ = 0.34) and Maintenance Planning Level (μ = 0.43). The deficiency is attributable to heavy vulcanized rubber build-up obliterating the transverse asphalt grooves.

1. ROOT-CAUSE ANALYSIS (WHY THE FRICTION DECAYED)

TECHNICAL ROOT CAUSE
Runway asphalt grooving completely clogged by 18,000 wide-body touchdown cycles over 14 months without hydro-vacuum de-rubberizing.

OPERATIONAL ROOT CAUSE
Maintenance intervals were scheduled on fixed calendar cycles (annual) rather than tracked dynamically by cumulative landing tonnage.

ORGANIZATIONAL ROOT CAUSE
De-rubberizing contractor contract expired in August without timely procurement renewal by airport procurement directorate.

2. THREE-STAGE CORRECTIVE ACTION PLAN (CAP)

PHASE	CORRECTIVE MEASURE	TARGET COMPLETION	VERIFICATION STANDARD
Phase 1 • Immediate	Issue permanent NOTAM warning of degraded braking action on RWY 14L under wet conditions. Prohibit tailwind landings exceeding 5 kt.	Within 12 Hours (Completed)	NOTAM published; AIS bulletin verified.
Phase 2 • Remediation	Deploy specialized ultra-high-pressure (UHP) hydro-blasting vehicles during night curfew (00:00–05:00) to strip rubber deposits.	Target: 25 Oct 2024 (14 Days)	Visual inspection; aggregate macro-texture depth ≥ 1.0mm.
Phase 3 • Validation	Conduct dual CFME calibration run across 3 runs (left, center, right of centerline) at 65 km/h and 95 km/h.	Target: 28 Oct 2024	Average friction coefficient μ ≥ 0.52; submit report to DGAC/EASA.

3. OPERATIONAL IMPACT & AUTHORITY ACCEPTANCE

The interim NOTAM ensures airlines apply factored wet landing distance calculations (+15% dispatch margin). The aerodrome operator commits to transitioning to a condition-based runway friction monitoring program where CFME runs occur every 60 days during high-traffic quarters.

FINDING STATEMENT • BLAST DEFLECTION FENCE DEGRADATION LEVEL 2 NON-COMPLIANCE

Inspection of Taxiway Papa holding point P1 identified 40 meters of curved jet blast deflection fence displaying structural rust corrosion at base anchor bolts, with 3 panels detached following high-power engine run-up tests. Additionally, the FOD containment wire mesh at the fence crest is torn in multiple locations, permitting gravel and ground debris to be sucked into adjacent engine intake corridors.

1. ENGINEERING & ORGANIZATIONAL ROOT CAUSE

STRUCTURAL & ENVIRONMENTAL FACTORS

Original fence specification in 2012 used standard galvanized steel rather than marine-grade epoxy coated anchors, leading to accelerated electrochemical corrosion from de-icing chemical runoff.

AVIATION TRAFFIC EVOLUTION

Stand 14 was reconfigured for wide-body A330neo and B787 operations producing breakaway jet blast velocities exceeding 120 kt, exceeding original 95 kt design limits.

2. CORRECTIVE ACTION PLAN & MILESTONE TRACKING

PHASE	ACTION ITEM	TARGET DATE	RESPONSIBLE UNIT
Milestone 1	Erect temporary heavy concrete jersey barriers with yellow reflective markers along perimeter service road.	Within 48 Hours	Airfield Civil Works Dept
Milestone 2	Install replacement curved corrugated blast panels with hot-dip galvanized Grade 316 stainless anchors rated for 150 kt exhaust.	Target: 15 Nov 2024	Infrastructure Contractor
Milestone 3	Install continuous high-tension nylon FOD catch-netting along top ridge of fence to trap windborne debris.	Target: 22 Nov 2024	Airside Maintenance Team
Milestone 4	Conduct physical pull-out torque test (45 kN per anchor) and joint authority site inspection for finding closure.	Target: 30 Nov 2024	Safety Manager + DGAC Inspector

3. TEMPORARY OPERATIONAL MITIGATIONS

Until Milestone 2 is completed, all aircraft departing Holding Point P1 must not exceed idle thrust until fully aligned with Taxiway Papa centerline. Ground vehicles on the adjacent perimeter road must hold at checkpoint Delta when aircraft are under power.

PEDAGOGICAL ARCHITECTURE • FLIPPED PEER-TEACHING: In Session 7, student teams step into the role of technical instructors. Each team of 3 to 4 engineers delivers a 30-minute communicative lesson to their classmates. The focus is authentic spoken communication, technical rigor, and interactive classroom facilitation.

1. RECOMMENDED TEACHING POD TOPIC SELECTIONS

DOMAIN 1: RUNWAY SURFACE CONDITION & GRF

Teach classmates how to assess runway friction decay, interpret continuous friction survey graphs, and draft accurate Global Reporting Format (GRF) SNOWTAMs.

DOMAIN 2: RAMP VEHICLE SAFETY & PUSHBACK

Facilitate a practical workshop on GSE speed enforcement, telematics geofencing, pushback towbar shear mechanics, and headset communication protocol.

DOMAIN 3: WILDLIFE HAZARD MANAGEMENT

Lead an interactive seminar on laser pyrotechnics, acoustic dispersal techniques, and habitat grass height regulation under EASA CS-ADR-DSN.

DOMAIN 4: AUDIT FINDING CLOSURE & CAPS

Run an interactive roleplay negotiation where classmates simulate aerodrome operators defending corrective milestones to civil aviation inspectors.

2. 30-MINUTE PEER-TEACHING LESSON ARCHITECTURE

STAGE	DURATION	COMMUNICATIVE GOALS & STUDENT ACTIONS
1. Opening Hook	~5 Minutes	Interactive briefing, photo exhibit analysis, or provocative technical scenario to engage classmates immediately in spoken English.
2. Core Activity	~18 Minutes	Student-centered simulation, information gap, roleplay, or safety case puzzle. The teaching pod circulates, coaches, and facilitates.
3. Debrief & Wrap-Up	~7 Minutes	Group presentations, peer feedback, technical vocabulary review, and error correction clinic conducted by the teaching team.

3. EVALUATION MATRIX • 20 POINTS TOTAL

1. PLANNING (5 PTS)

Clear objectives, logical pacing, polished student handouts, and balanced workload across all team members.

2. ACTIVITY (5 PTS)

High student talking time (STT), authentic technical challenge, interactive problem solving, creative ideas.

3. SPOKEN ENGLISH (5 PTS)

Fluency, natural communicative instructions, accurate technical aviation terminology, clear pronunciation.

4. FACILITATION (5 PTS)

Classroom presence, engaging peer interaction, dynamic question handling, constructive error correction.